LegalSpan header attention and CRF

CS781 Phase 2 legal named entity recognition | source-audited report

Task and evidence

Predict exact original-character entity spans in 12 legal categories. The source archive holds the actual code, configuration, run record, selected weights, and prediction ZIP. This report was produced after training from those records; it is not an executed historical notebook.

Architecture

The selected LegalSpan LoRMA NER checkpoint is merged into the backbone, then fully tuned. A header-character BiGRU and four-head attention inform each character; a 25-tag CRF and boundary loss encourage coherent spans.

Path	Legal text + header -> Merged LegalSpan -> Header BiGRU -> 4-head cross-attention
Output	BIO emissions + CRF -> Boundary loss -> Span decoder

Training and validation

Method: Full tuning + header attention + CRF. This export's archived held-out strict macro-F1 is 0.411516 on the 12-document local validation split. Its selected checkpoint and prediction ZIP hashes are in evidence/run_summary.json. The official test set has 35 document IDs; no hidden Codabench score is established by this archive.

Implementation map

Key original files: boost_model.py, run_boost.py, vendor/lner/models.py. The companion notebook explains the tensor route and reads archived evidence without presenting invented execution outputs.

Reproducibility boundary

The package includes exact source snapshots and small run records, plus the archived prediction ZIP. Large model weights, tokenizers, and course data stay in the separate complete archive. Original machine paths in config files need remapping for a new environment. The consensus rule was selected on validation; 100M/200M D222 exports were completed after the original deadline where marked.

Interpretation

Model variants change either pretrained backbone, parameter-update method, NER head, or pretrained checkpoint. A score difference does not isolate one factor unless all other choices match. The local held-out set is small, so its ranking should not be described as a hidden-test result.

Recorded model and adaptation

The source and run records together identify the following settings. Counts include the Phase 2 NER head. The method changes which backbone parameters are updated; it does not change the original Phase 1 tokenizer during this run.

Field	Archived value
Base family	LegalSpan
Adaptation	Full tuning + header attention + CRF
Total parameters	112991117
Trainable parameters	112991117
Backbone LR	1e-05
Head LR	0.0002
Pretrained SHA-256	2eb64c8489aa4ead38ec…
Selected NER SHA-256	2cb1c2dbb702b9ef85b9…

Training signal and exact-span evaluation

For the common NER runner, the character tagger minimizes weighted cross-entropy over 25 BIO categories; A0 also matches its verified B1 teacher with temperature-2 KL. D222 adds CRF negative log-likelihood and a weighted boundary objective. HeaderCRF combines cross-entropy, CRF, and boundary losses. The validation scorer counts a prediction as correct only when the start, end, and label all match the gold span; macro-F1 averages the 12 per-label F1 values equally.

Observed optimization trace

These five entries are sampled from the archived training log; loss is a per-batch training value, not held-out macro-F1. Its decline alone cannot establish improved entity extraction.

Optimizer step	Epoch	Logged batch loss
1	1	0.4442
210	2	0.2569
420	3	0.2686
630	4	0.2221
830	5	0.2137

Label-level held-out scores

The following F1 values come from the archived selected validation prediction. Low-frequency labels are sensitive to a few documents, so read them beside their support and error counts in evidence/selection.json.

Entity label	F1	TP	FP	FN
APP	0.110	10	62	99
RESP	0.048	4	60	97
A.COUNSEL	0.505	24	26	21

Entity label	F1	TP	FP	FN
R.COUNSEL	0.416	16	30	15
JUDGE	0.594	19	10	16
WIT	0.000	0	32	21
AUTH	0.222	44	147	161
COURT	0.719	150	57	60
STAT	0.519	148	78	196
PREC	0.281	80	227	182
DATE	0.939	313	22	19
CASENO	0.584	80	56	58

Verification and reproducibility

The prediction ZIP has a single root prediction.json, passed CRC, and contains exactly 35 official test IDs. Its SHA-256 was compared with the completed run record. Each copied code file has a SHA-256 listed in evidence/SOURCE_HASHES.json. Exact re-execution requires the original selected weights, tokenizer and course data preserved in the complete archive; this compact package does not embed them or assert that historical notebook output cells existed.